COMPUTER AND COMMUNICATION TECHNOLOGY

INTRODUCTION TO COMPUTER NETWORKS

TYPES AND SIZES OF NETWORKS

- The three most common types of networks :
  - Local Area Network (LAN)
  - Metropolitan Area Network (MAN) and
  - Wide Area Network (WAN)
- WAN connects sites that are in diverse locations.
- LAN connects machines within a site, for example, a university campus.
- In a university, a single department or college has its own LAN.
- The department or college LANs are connected to university LAN or MAN.
- The university LAN or MAN is connected to WAN via leased lines which are private lines providing permanent pathway between two communicating stations.
- Another example is an enterprise or corporate network.
- It may consist of multiple LANs that may be connected over a distance to form a WAN.

Wide Area Network (WAN)

- A WAN usually refers to a network that covers a large geographical area and uses common carrier circuit to connect intermediate nodes.
- The WAN for a multinational company may be global whereas WAN for a small company may cover only a few cities.
- A major factor that distinguishes WAN is that it utilized circuits leased from telephone companies or common carrier.
- This restricts the communication facilities and transmission speeds to those provided by such companies.
- Transmission rates typically ranges from kbps to Mbps with 56kbps, 64kbps, 2Mbps, 34 Mbps and 45 Mbps.

Metropolitan Area Network (MAN)

- A MAN covers an area between 5 and 50 km, about the size of a city.
- It acts as a high speed network to allow sharing of regional resources (similar to a large LAN).
- A MAN like WAN is generally not owned by a single organization, but rather a consortium of users or by a single network provider who sells the services to users.
- The level of service provided to each user must therefore be negotiated with the MAN operator and some performance guarantees are normally specified.
- Its primary customers are companies that need a lot of high speed digital services within relatively small geographical area.
- A man is also frequently used to provide shared connection to other networks using a link to WAN.

Local Area Network (LAN)

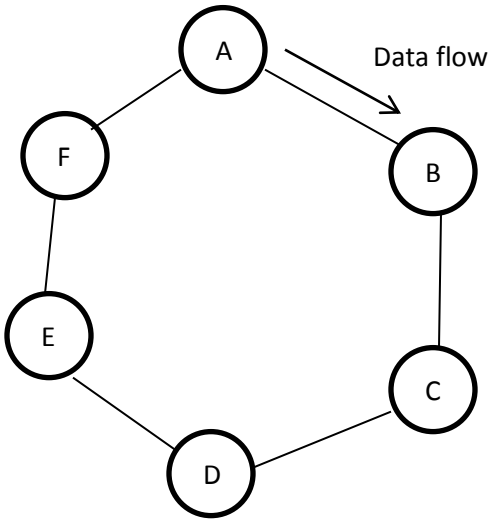
- LAN is the most common type of data network.
- Typical installations are :
    - Industrial plants
    - Office buildings
    - Colleges or university campuses or
    - Similar locations
- LANs are installed by organisations that want their own high quality, high speed communication links where data speed range from 10 Mbps to 1000 Mbps.
- LAN allows users to users to share computer related resources within an organistation band may be used to provide shared access to remote users through arouter connected to a MAN or WAN.
- Intermediate node devices such as repeaters, bridges, switches and routers allow LANs to be connected together to form large LANs.
- At local level, a backbone is a line or a set of lines that LANs connect to for a WAN connection or within LAN to span distances efficiently – for example, between buildings.

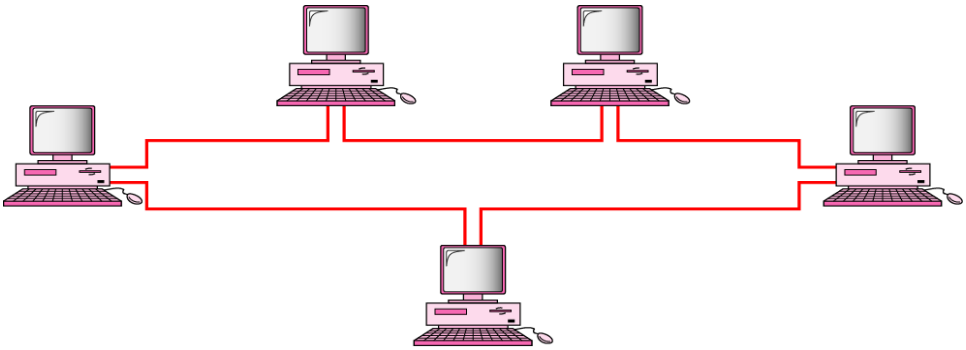
TOPOLOGIES

- A topology gives a physical layout.
- In communication networks, it describes configuration or arrangement of network including nodes and connecting lines.
- The ring bus and star are the basic network topologies.

Ring Topology

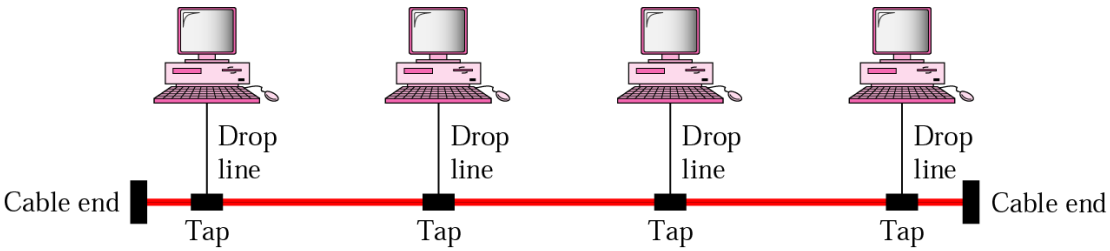
- Ring is a network topology or circuit arrangement in which each device is attached along the same signal path to two other devices and form a path in shape of a ring.
- Each device in the ring has unique address.
- To avoid collisions, information flow is unidirectional and controlling device intercepts and manages the flow to and from every station on the ring by granting a token or permission to send or receive.
- The advantage of a ring is that it is easy and inexpensive to install and if one connection is down, the network will still work.
- The disadvantage is that:
    - the network must be shut down for reconfiguration
    - It is difficult to trouble shoot.
- Examples of a ring network are:
    - Token ring,
    - Fibre Distribution Data Interface (FDDI)
    - Synchronous optical network (SONET)





Bus Topology

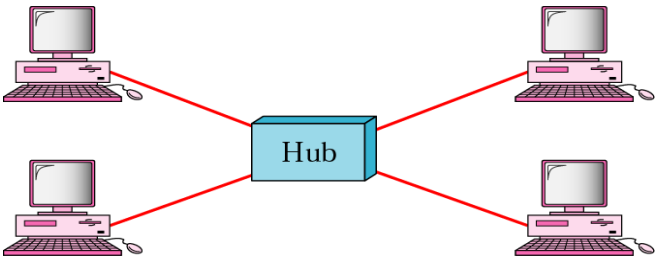
- Bus topology is a circuit arrangement in which all devices are directly attached to a line and the signal passes through each of the devices.
- Each device has its own unique identity, and can recognize those signals intended for it.
- The advantages of the bus are:
  - It is inexpensive
  - It is simple and easy to configure, connect to and expand.
- The disadvantages are:
  - that if the backbone goes down the whole network goes down
  - Increasing the number of users will cause the network to become slower and the performance may be unpredictable under heavy load condition.
  - The network must be shutdown to add any new users.
  - Trouble shooting can be time consuming.
- Examples include 10base2, thin Ethernet.



Star Topology

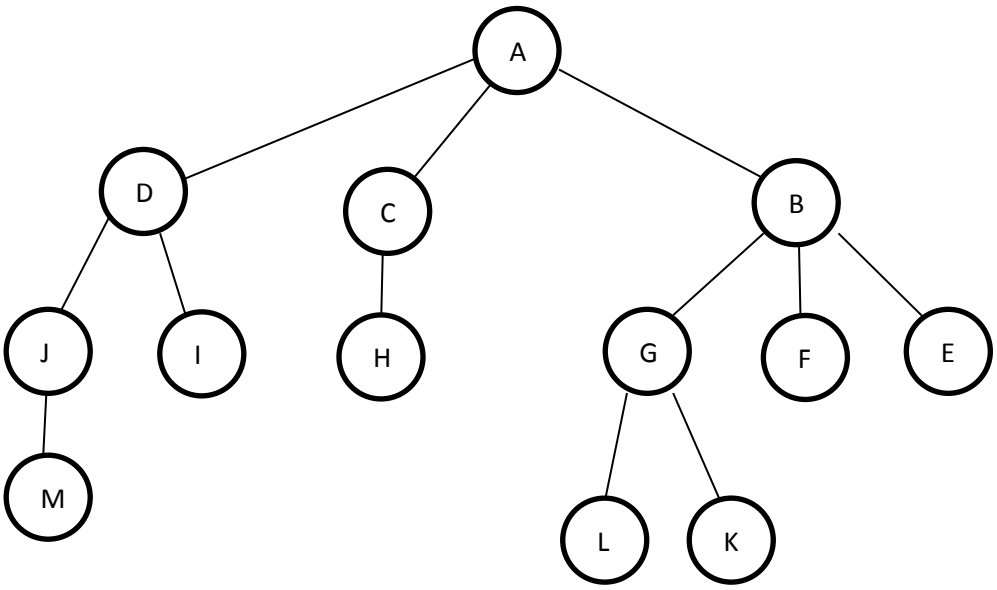
- Star network is a network in which all devices are connected to a central node.
- The node rebroadcasts all transmission received from any peripheral to all peripheral nodes on the network including the original node.

- Thus all peripheral nodes may communicate with all other nodes by transmitting and receiving from the central node.
- Advantages:
    - It is simple and robust
    - It is faster than ring or bus
    - Has greater stability
    - It is easy to setup, reconfigure and trouble shoot.
    - Has low configuration cost
    - Provides centralized administration and security control.
    - If the transmission line linking a peripheral node to central node fails, it will result in isolation of that peripheral node but the remaining network is not affected.
- Disadvantages:
    - If one of the hubs fail, or a hub cable fails, it will shut down that segment of the network.
    - A hub has a limited number of ports and an increase in the number of users may involve additional network expenses.
- Examples include 10BaseT, 100BaseT, 1000BaseT Ethernet.



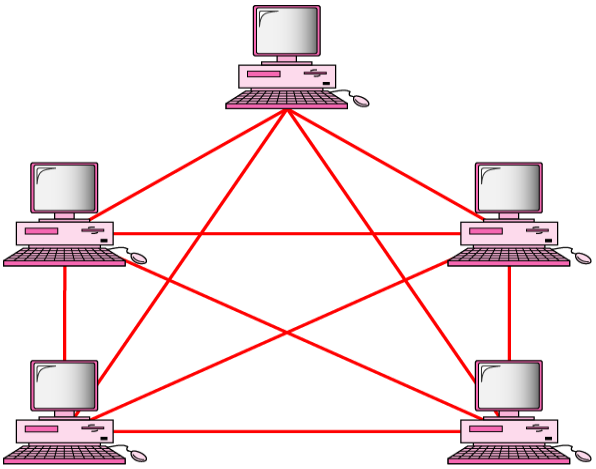
Tree Topology

- Tree topology is also known as hierarchical network, is a network topology that purely forms a topological viewpoint that resembles star network
- The individual peripheral nodes are required to transmit to and receive from one other node only, towards the central node, and are not required to act as repeaters or regenerators.
- The function of the central node may be distributed.
- The top node in the structure is called the root node.
- The advantage is that it is to add more components
- The disadvantage is that like star, if the central node fails, the whole system fails.



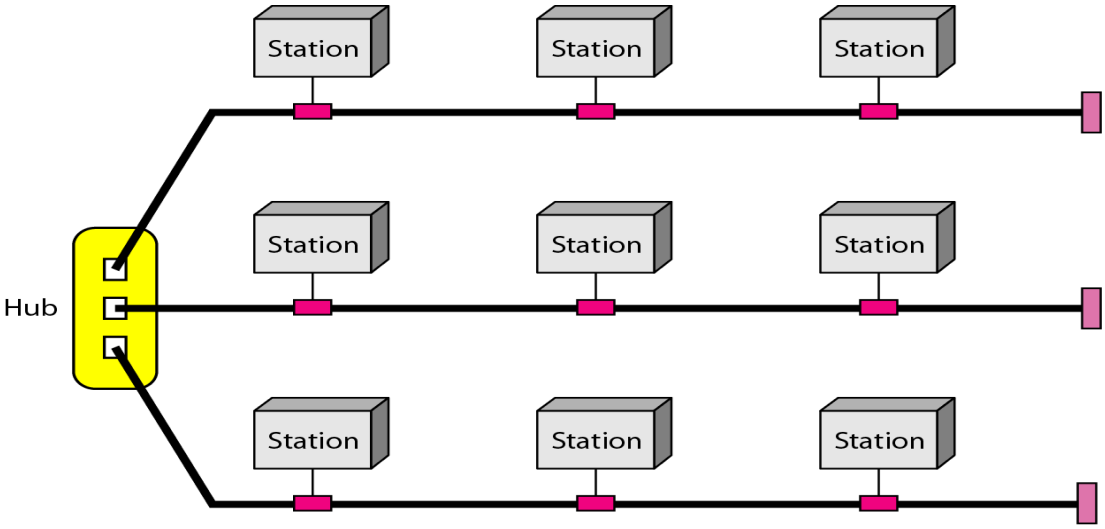
Mesh Topology

- This is similar to hierarchical structure except that there are more interconnections between nodes at different levels, or even at the same level.
- At a minimum there are at least two nodes with two or more paths between them.
- In a fully interconnected mesh, each node is connected to every other node, although this is cost prohibitive and therefore seldom implemented.
- The PSTN (Public Switched Telephone Network) is a classic example of mesh topology with multiple interconnections making the network virtually fail safe.
- LAN backbone switches may be connected using mesh topology.



Hybrid Topology

- This is a combination of two or more basic network topologies.
- Instances where two basic network topologies and they retain the basic network character cannot be described as hybrid network.
- For example, a tree network connected to a tree network will still be a tree network.
- Therefore, a hybrid network is only created when two basic network topologies are connected and the resulting network topology fails to meet one of the basic topology definitions.
- For example ring and star network connected together exhibit hybrid network topology.



CHARACTERISTICS OF COMMUNICATION NETWORKS

- The characteristics used to differentiate between LAN, MAN and WAN are listed below:

1. *Geography*

- LAN is restricted to a limited geographical coverage of a few kilometres, typically less than 5Km.
- MAN can cover between 5km and 50 km
- WAN may cover more than 50 km.

2. *Ownership*

- A LAN is owned by a single organization because of limited geographical area.
- WAN is formed by interconnection of multiple LANs each of which may belong to a different organization.
- Therefore administrative complexities and cost of LAN are usually much lower than WANs.
- MAN may be owned by one organization or different organisations.

3. *Data Rates*

- Data rates are higher in LAN usually from Mbps to Gbps.
- The data rates in MAN are in Kbps to Mbps.
- The data rates in WAN range from Kbps to Mbps.

4. *Error Rates*

- LAN generally experience fewer data transmission errors than WAN.
- Typical errors in LAN are 10^{-8} to 10^{-7}.
- Typical errors in WAN are 10^{-5} to 10^{-7}.

5. *Typical Applications*

- LAN
    - Industrial plant
    - Business offices
    - College campuses
    - Single department
- WAN
    - Connect offices in different cities using leased lines

- MAN
    - Frequently provide shared connection to other networks using a link to WAN.
    - City networks.

6. *Communication Link*

- LAN uses twisted pair cables, coaxial cables and fibre optic cables.
- WAN uses telephone lines, microwave link and satellite channels.

7. *Communication Cost*

- The overall communication cost of LAN is much lower than WAN.
- The main reasons are:
    - Lower error rates.
    - Simple algorithms.
    - Lower administrative and maintenance cost.
- The cost to transmit data in LANs is negligible since the transmission medium is owned by the user organization.
- However, with WAN, this cost may be very high because the transmission media used are leased lines or public communication system such as telephone lines, microwave links or satellite channel.

SYNCHRONOUS AND ASYNCHRONOUS TRANSMISSION

- Synchronous and asynchronous data transmission is used with serial data transmission.

1. *Synchronous Transmission*

- In synchronous systems, separate channels are used to transmit data and timing information.
- The timing channel transmits the click pulses.
- Upon receipt of the clock pulse, the receiver reads the data channel and latches the bit value found on the channel at that moment.
- The data channel is not read again until the next clock pulse arrives.
- Because the transmitter originates both data and timing pulses, the receiver will read the data channel only when told to do so by the transmitter and synchronization is guaranteed.

- Techniques exist to merge the timing signal with the data so that only a single channel is required.
- This is especially useful when synchronous transmission is to be sent through a modem.

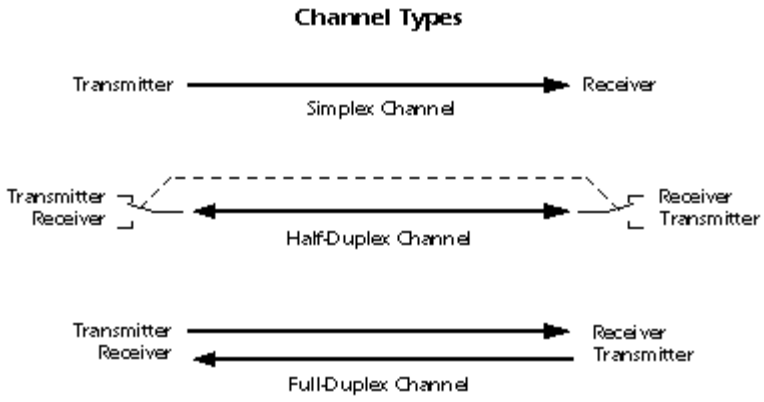
2. *Asynchronous Transmission*

- In asynchronous transmission, a separate timing channel is not used.
- The transmitter and the receiver must be preset in advance to an agreed upon baud rate.
- A very accurate local oscillator within the receiver will then generate an internal clock that is equal to the transmitter's within a fraction of a percentage.
- For most common serial protocol, data is sent in small packets of 10 to 11 bits of which 8 constitute message information.
- When the channel is idle, the signal voltage corresponds to a continuous logic "1".
- Data packets always begin with logic "0" (the start bit) to signal to the receiver that transmission is starting.
- The start bit triggers an internal timer in the receiver that generates the needed clock pulses.
- Following the start bit, 8 bits of the message data are sent bit by bit at the agreed upon baud rate.
- The packet is concluded by a parity bit and a stop bit.
- The packet length is shorter in asynchronous system to minimize the risk that the local oscillator in the receiver and transmitter will drift apart.

DIRECTION OF TRANSMISSION FLOW

1. *Simplex*

- The message source is the transmitter.
- The message destination is the receiver.
- A channel whose direction of transmission is unchanging is referred to as a simplex channel.
- For example a radio station.



2. *Half Duplex*

- This is a single physical channel in which direction may be reversed.
- Messages may flow in two directions but never at the same time.
- In a telephone call one party speaks while the other party listens.
- After a pause, the other party speaks and the first party listens.
- Speaking simultaneously results in garbled sound that cannot be understood.

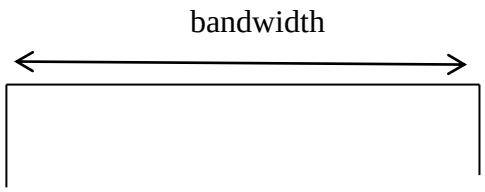
3. *Full Duplex*

- Allows simultaneous message exchange in both directions.
- It really consists of two simplex channels, a forward channel and a reverse channel linking the same points.

TRANSMISSION RATE AND BANDWIDTH

- Bandwidth is the range of frequencies that can be transmitted with minimum distortion.
- The bandwidth is equal to the rate of information transfer or transmission rate, which is equal to amount of information that is communicated from the source to the destination in a fixed amount of time typically one second.
- Bandwidth is also a measure of transmission capacity of a communication medium.
- There is a general rule that relates bandwidth and information capacity.
- Hartley's Law, which states that the amount of information that can be transmitted in a given time is directly proportional to the bandwidth and is represented by the equation below.
- I = KTB where
    - I is the amount of information that can be transmitted.
    - K is a constant that depends on the type of modulation.
    - T is the transmission time in seconds.
    - B is the channel bandwidth
- From the above equation, it is clear that the greater the channel bandwidth the greater the amount of information you can transmit in a given time.
- You can still transmit the same information over a narrower channel except that it will take longer.
- High bandwidth applications include:
    - Web browsing,
    - E-commerce
    - Audio and video streaming
    - Real time document sharing
    - Video conferencing
    - On line gaming
    - Digital TV
- As the movement for transmission of data, voice and video traffic over the same network continues to gain momentum, the demand for bandwidth keeps growing.
- For digital devices, the bandwidth is expressed as bits per second (bps).
- In most cases, bandwidth is the same as channel frequency, so 100 MHz is analogous to 100 Mbps.
- For analog devices, bandwidth is expressed in cycles per second or Hertz (Hz).
- The minimum channel bandwidth is determined by the difference between the upper frequency and lower frequency limits of the signal.

- For example, since most humans speech falls in the frequency range of 200Hz – 3000Hz, the minimum bandwidth requirement is 2800Hz but 4000 is allocated.

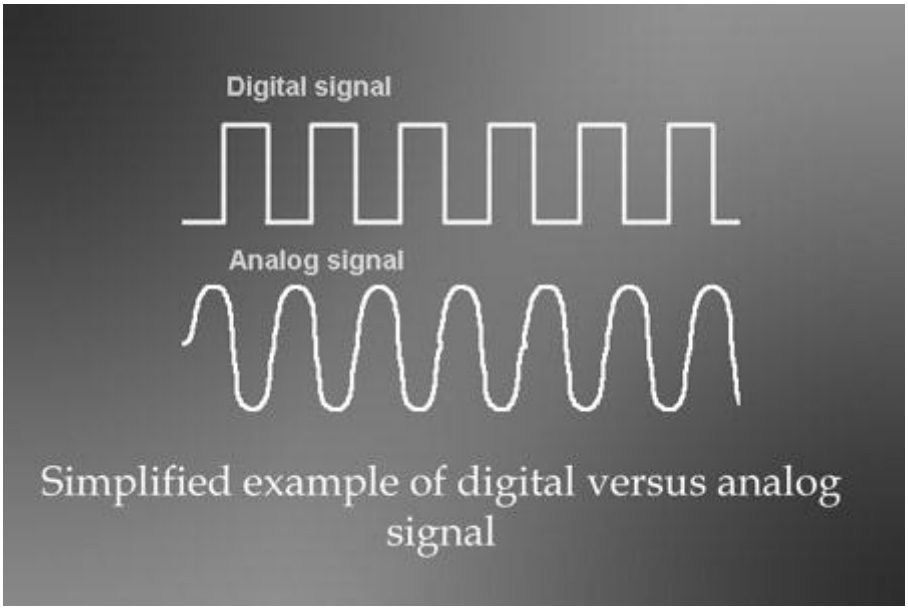


f1 = 200Hz f2 = 3000Hz

lower frequency limit upper frequency limit

TYPES OF SIGNALS AND STANDARDS

- Information that need to be communicated can be in analog or digital form.
- Analog signals are continuous varying quantities.
- Digital signals are discrete quantities, most commonly binary (On or Off, High or Low, 0 or 1).
- Voices, images and temperature readings from a sensor are all examples of analog data.
- In digital transmission, all information is reduced to a stream of 0s and 1s, you can use single network for voice, data and video.
- Digital circuits:
    - Are cheaper.
    - Are more accurate
    - Are more reliable
    - Have fewer transmission errors
    - Are easier to maintain
- than analog circuits.
- Vast structures exist for analog signaling and much of it can be adapted to transport digital signals as well.
- The public telephone network, cable TV infrastructure and partially every form of wireless communication are inherently analog transmission media that have been adapted for digital transmission.



- Analog data can be encoded as an analog signal, for example, cassette tape player and audio video component of TV program.
- Digital data is regularly represented by digital signal, for example, email.
- Also, analog data is also commonly encoded with digital signals.
- When you scan an image or capture sound on the computer, you are converting analog data to digital signal.

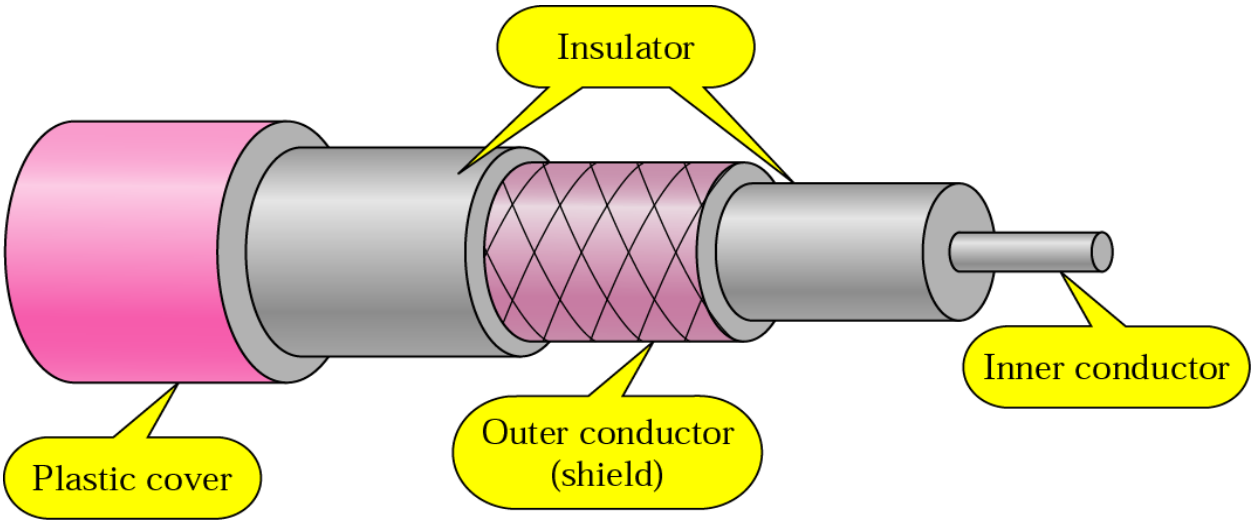
- Digital transmission has replaced analog transmission in PSTN, except the telephone local loop, which is a pair of copper wires that run from a telephone to a local switching station.
- Although voice is the primary signal carried by the local loop, this network is now widely used to carry digital information or data as well.
- A modem is used on local loop to convert digital signals to analog for transmission.
- A modem also converts analog signal received from a local loop into digital form.

TRANSMISSION MEDIA

- A transmission medium is a technology used to carry signals from one device to another across the network.
- The choice of media depends on:
    - Cost
    - Quality
    - Speed and
    - Range of data travel.
- The transmission media can be grouped into two categories:
    - Guided media and
    - Unguided media
- Example of guided media include:
    - Coaxial cable
    - UTP (unshielded twisted pair cables)
    - STP (Shielded twisted pair cables)
    - Fibre optic
- Examples of unguided media include:
    - Infra-red
    - Bluetooth
    - Radio waves (electromagnetic waves)

Coaxial Cable

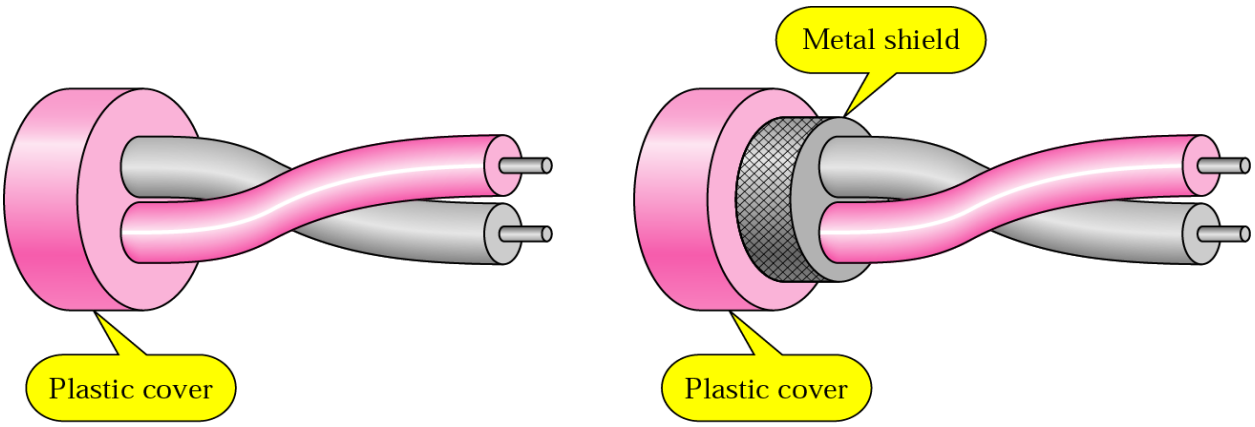
- This is an older method of cabling networks.
- Although it was a networking medium of choice in the late 1980s and early 1990s, it was rapidly replaced by UTP (untwisted pair cables)
- However, more recently, coaxial cabling has made a comeback with home broadband/ADSL services and has always had a place with terrestrial TV (in a rooftop aerial).
- Coaxial cable is so named because it has copper core surrounded by plastic sheath and then another copper braid.
- The inner cable is used to transmit data and the outer cable (usually interlaced braid of copper wires) is used as ground (connects to earth).



UTP (Unshielded twisted pair) and STP (Shielded twisted pair)

- UTP is the most popular cable in use on academic and commercial networks throughout the world.
- UTP and STP originated during a change in phone technology in 1980s and have endured many revisions.
- UTP and STP are still in use for many reasons:
    - They are low cost in comparison with fibre optic and coaxial cable
    - They are very versatile and can be adapted to many uses.
    - They can be adapted to changing speeds and standards.

- UTP comes in long rolls that contain 8 cables inside one jacket; these cables come in coloured pairs.
- Each cable generates small magnetic field which will create interference for the neighbouring cables.
- Each pair is twisted around each other to reduce effect called cancellation, which occurs because of proximity of cables, hence the term twisted pair.
- Four pairs are used.



a. UTP b. STP

- UTP is susceptible to interference and noise.
- With no protection, external sources such as power supplies or cables can also corrupt data sent along the media.
- To overcome this, more expensive STP can be used.
- It operates on the same principles as UTP, but has extra foil wrapping around the cables for protection.

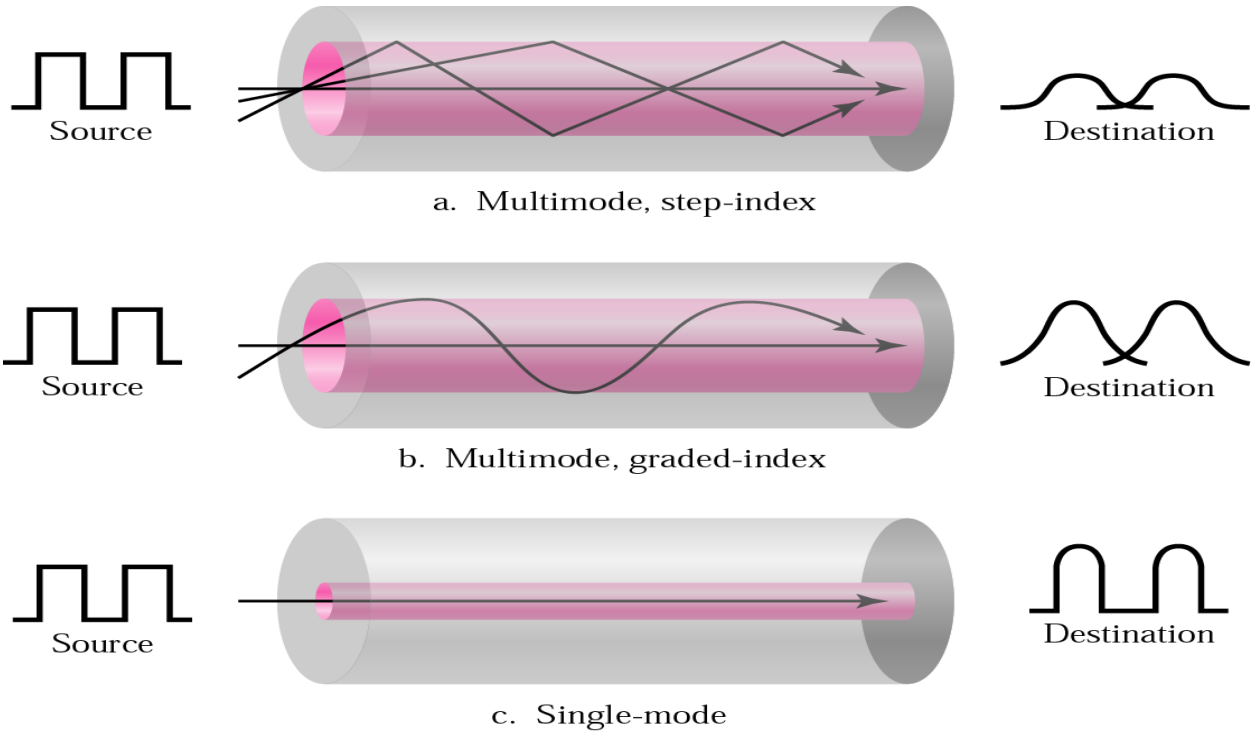
- UTP is rated in categories.

- The current LAN standards include: categories 5, 5e, 6 and 7.
- Categories 1, 2, 3, and 4 (including coaxial cable) are no longer used for LANs and are found solely in the domain of WAN Technology (as WAN often use slower connections between buildings).
- Each category is related to the quality of the cable, its connectors and the speed at which data can be sent reliably.
- With category 6 and 7 cables, all the termination (wiring) is done within the factory or with a specialist equipment.
- Category 5e is normally terminated with an RJ45 plug or socket and can be easily be done with low cost equipment.

Fibre Optic

- Both coaxial cable and UTP/STP rely on the transmission of data using electrical pulses down the copper cable.
- Fibre optic uses light to transmit data and so can achieve greater speed and distance than their copper counter parts.

- There are two type of cables in common use:
  - Single mode, which is 8 microns in diameter.
  - Multimode, which is 125 microns in diameter (with inner core of 62.5 microns).



Multimode Fibre

- The light source for multimode fibre is the light emitting diode (LED)
- An LED sends frequent pulses of light down the cable (no pulse equal to digit 0, a pulse equal to digit 1).
- The light will travel around the cable based on the different densities of the medium it is travelling through.
- As it reaches denser material, most of the light will be reflected and will continue its path in the cable.
- Some of the light will be refracted (absorbed) into the denser outer coating.
- Overall, this allows the light to travel up to three kilometres before becoming too weak.
- On a LAN, a cable normally runs 285 metres guaranteeing, a signal that can carry data rates up to 1 gigabits per second.

Single Mode Fibre

- Single mode fibre optic cable is thinner (8 microns) and uses class 3 laser to send the data.
- The smaller diameter reduces an angle of reflection and therefore minimizes refraction and loss.

- A class 3 laser is dangerous and is outside the visible light spectrum
- If you accidentally look into class 3 enabled fibre optic cable, you will not see any light but will burn the cells at the back of your eye.
- This will cause a permanent blind spot in your vision.
- Fibre optic cable is more costly than copper counter parts.
- The termination of the cable has to be done under exacting (requiring precise accuracy) and clean conditions.

Unguided Media

1. Infrared
    - Used by mobile phones PDAs, printers and laptops as a method of local peer communication.
    - Infrared is on electromagnetic spectrum, slightly below the visible light.
    - For two devices to communicate they need line of sight.
2. Bluetooth
    - Is available in many systems.
    - It uses utra high frequency spread spectrum signal.
    - Bluetooth uses a range of frequencies and techniques to send data.
    - The purpose of Bluetooth is to create mobile short range networking for small devices.
3. Radio (wireless)
    - Radio waves and microwaves operate between 3 KHz and 300GHz.
    - Radio waves are used for multicast communication such as radio, television and paging systems.
    - Microwaves are used for unicast communication such as cellular telephones, satellite networks and wireless LANs.